

# Uncovering Global Music Patterns Through Spotify Data

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Table 1: Group Contribution

| Member Name   | Contribution                                                         |
|---------------|----------------------------------------------------------------------|
| Danny Kuei    | Data fetching from MusicBrainz, Country trend analysis               |
| Chi-Chun Chen | Web-Scraping, Data Fetching from Spotify API, Lyrical Analysis       |
| Liang-Yin Tao | Web-Scraping, Data Fetching from Spotify API, Top 50 Charts Analysis |

## Abstract

Streaming data provides a valuable lens for understanding global music preferences. This study analyzes Spotify chart data to investigate how listening trends vary across countries and genres. Webscraping was utilized to obtain weekly and total Spotify chart data including streams, artist, and song while the MusicBrainz API matched the artist to country of origin and genre. Additional song characteristics, including duration and release date were collected from the Spotify API while lyrics were retrieved from the Genius API.

We analyze geographic as well as country level patterns. Countries vary heavily in streaming domestic artists versus international artists. Results also indicate that while the United States (US) and English speaking nations dominated historical charts, recent trends show increasing influence in Spanish-speaking Latin American countries. Likewise, while pop and hip-hop are the two most widespread genres worldwide, reggaeton music has emerged as a major genre. Analysis of the top songs ranking suggest song duration varies slightly across country but is not predictive of rank. Lyrical analysis using Latent Dirichlet Allocation (LDA), sentiment analysis, and keyword analysis reveals distinct themes among top songs in the US.

# 1 Introduction

While streaming numbers are not indicative of artist success, in the current digital landscape, they provide valuable information about artist popularity and cultural influence. Not only can large artists earn royalties from streams, small artists now have new avenues to gain digital exposure (1). Since its launch in 2006, Spotify has expanded to 184 markets and has become a major music platform. However, the extent to which Spotify operates as a global platform remains an open question.

Historically, the music industry has experienced major shifts in genres and trends, with rock and roll dominating the 1950s and hip-hop and pop surging in the late 1990s.(2). At the same time, the music industry has long been dictated by English-speaking countries such as the US, United Kingdom (UK), and Canada, as reflected in history and global album sales (3). This raises the question of whether modern streaming platforms reinforce this or reflect more diverse global listening patterns.

In this study, we analyze user listening preferences across countries using Spotify streaming data. Our dataset consists of weekly Spotify charts containing the top 20 streamed songs from 76 countries for a total of 1,540 songs. This was combined with artist metadata of country and genre, to allow for cross-country comparisons. From looking at historical totals and comparing vs the weekly streams data, we can examine if Spotify streams reflect a globalized music market or remain concentrated around a small set of countries or genres.

In addition, we also investigate characteristics of popular songs. Recent reports, such as a notable piece by *The Economist* (4), have highlighted a growing industry trend where popular songs are becoming increasingly shorter. Prompted by this observation, we sought to investigate whether shorter song durations actually correlate with higher chart rankings as shown in Spotify streaming charts and whether duration varies across countries. Finally, we analyze lyrical content of top songs in the US to identify thematic trends.

## 2 Methods

### 2.1 Data Acquisition and Dataframe Subsets

To obtain Spotify charts, we performed webscraping using the BeautifulSoup package in Python on Kwordb.net. Kwordb is a website that provides charts for a variety of streaming platforms such as Spotify, Youtube Music, and iTunes. The charts include daily, weekly, and all time charts for both global and country data with 76 countries represented for Spotify.

For global comparisons, we were interested in looking at artist info such as country of origin and main genre. To obtain this information, we used MusicBrainz.org, an open source database containing artist, album, and track information. MusicBrainz has an API which was used to match artists from the chart to country and genre. Due to rate limits, a limited number of songs were scraped and fetched. When looking at all 76 countries, only

the top 20 weekly songs for each country were scraped. For all time global charts, the top 500 artists were scraped. Genre being user community inputted led to a lot of missing data especially from artists from non English speaking nations. Country data was also missing for some artists or included a city or state they were based in. Manual imputation was thus performed on the artists that appeared on charts multiple times. Around 26.0% of artists were missing country with 18.4% missing post imputation while genre had 51.6% artists were missing with 45.4% missing post imputation.

For Top 50 Charts Analysis, the Spotify API was utilized to extract deeper insights from the charts obtained in the previous step. We retrieved detailed track-level information, specifically focusing on track duration, explicit content labels, and release dates. Due to time constraints and the need to respect API rate limits, we restricted this phase of the analysis to a representative subset of the data. 50 song requests were made for each of the 10 charts we made comparisons for. Our selection features the top five markets by Spotify streams—the US, Mexico, Indonesia, Brazil, and India—alongside the Global chart for a baseline comparison. We subsequently added Italy, Austria, Japan, and Taiwan to ensure greater regional diversity across Europe and Asia, as well as to reflect personal interests and high user engagement in those countries.

For lyrical analysis, song lyrics were collected from [Genius.com](https://www.genius.com) using the Genius API. Genius provides publicly available lyrics and metadata for a wide range of songs. Due to time constraints and the need for linguistic consistency, our analysis focused on the Spotify US Top 200 chart, from which lyrics for 198 songs were successfully retrieved and matched with the chart data. Songs with missing or unmatched lyrics were excluded from the lyrical analysis dataset. For songs with lyrics in languages other than English, we used the English translated version when a translation was available on Genius, allowing the corpus to be standardized for downstream NLP analysis.

## 2.2 Top 50 Charts Analysis

To analyze the characteristics of top-performing tracks, we focused on three primary dimensions: song duration, explicit content, and release dates. For song duration, we employed data visualization techniques to compare the distribution of track lengths across the selected countries. This allowed us to identify whether certain markets exhibit a preference for shorter tracks or longer compositions. Regarding explicit content, we calculated the proportion of songs labeled with "adult content" on each chart to determine if cultural or regional factors influence the prevalence of explicit lyrics in mainstream hits. For release date analysis, we converted the raw data into "song age" to investigate consumption patterns, specifically exploring whether certain countries show a stronger inclination toward nostalgic older tracks or if the charts are predominantly driven by fresh new releases.

## 2.3 Lyrics Analysis

For lyrical analysis, we applied several natural language processing (NLP) techniques to examine thematic and emotional patterns in popular music lyrics.

Before analysis, the lyrics were preprocessed to standardize the textual data. This pre-processing pipeline included converting all text to lowercase, removing punctuation and special characters, eliminating structural labels such as “[Verse]” and “[Chorus],” and removing common stop words and other non-informative words. The cleaned lyrics were then tokenized into individual words to construct a document-term representation for each song.

First, we applied Latent Dirichlet Allocation (LDA) topic modeling to uncover latent thematic structures in the lyrics (5). In this framework, each song lyric was treated as a document, where each document is assumed to be a mixture of multiple topics, and each topic is represented by a distribution of words. By examining the most representative words within each topic, we identified recurring lyrical themes across the most streamed songs. The LDA model also generated a topic distribution for each song, and the topic with the highest probability was assigned as the song’s primary topic for subsequent analysis. Topic interpretation and visualization were further supported using LDAvis (7).

To further examine emotional patterns, we conducted sentiment analysis using the VADER sentiment lexicon (6). VADER assigns sentiment scores to text based on a predefined lexicon and produces a compound sentiment score ranging from -1 (most negative) to 1 (most positive). This allowed us to evaluate the overall emotional tone of songs within the dataset.

Finally, we performed keyword and phrase frequency analysis by examining common unigrams and n-grams to identify frequently occurring linguistic patterns in popular music lyrics.

## 2.4 Visualizations

All visualizations were created using Python libraries. Maps were made using folium as well as seaborn for the genre map. Country names were converted to two letter country code using the ISO 3166-1 alpha-2 definition which required the utilization of the pycountry package. Additional visualization libraries included Plotly for barplots and PyVis for interactive web plots.

# 3 Results

## 3.1 Global Streaming Trends

To begin, we analyzed overall streaming patterns across all countries. As expected the US had the highest total stream counts (Figure 1). The next nine countries were then Mexico, Indonesia, Brazil, India, Italy, Turkey, UK, and France. These results highlight Spotify’s

broad international reach in Europe and Latin America. However, the map shows that data was limited in most of Africa and many parts of Asia.

The top 500 artists showed much less geographic diversity with only around 30 countries represented (Figure 2). English-speaking artists dominated the rankings the US contributing 245 artists and UK contributing 58 artists, totaling around 60% of the list. However, Latin American countries were also well represented with Puerto Rico, Mexico, Colombia, and Brazil all having more than 10 artists in the list. India, Canada, South Korea, and Australia also had a significant number of artists represented.

Genres distributions among the top 500 artists revealed that hip-hop and pop were the two most dominant genres (Figure 2). After these two categories, there was a notable drop-off in frequency before Reggaeton and Latin music following behind. Additionally Latin genres such as Regional Mexicano, trap Latino, and Latin pop were also prominent. Other genres that had moderate frequency were rock, R&B, country, K-pop, and electronic music.

However, examining the dominant genre in recent charts revealed distinct regional patterns (Figure 2). Reggaeton appeared not only throughout Latin America but was also prevalent in the US, Spain, and Italy. In general though, European countries tended to prefer hip-hop and pop, while Asian countries showcased a higher diversity of genres. In particular, they favored regional genres such as J-rock in Japan, K-pop in South Korea, Filmi in India and Pakistan, and Indonesian pop in Indonesia.

Next, we looked at if countries preferred to stream either domestic or international artists (Figure 3). The countries with the highest domestic shares included Israel, Hungary, South Korea, Indonesia, Egypt, Finland, and India. This pattern could reflect stronger local music industries, cultural preferences for domestic artists, or differences in Spotify user demographics. In contrast, the countries with the highest shares of international artists streamed included several Latin American countries. This could be explained by being able to listen to neighboring Spanish speaking artist. Furthermore, this suggests that users prefer listening to music in their own language which could explain the high domestic percentage of the aforementioned countries with have the language being endemic. Notable exceptions in Latin America included Brazil where Portuguese is spoken, and Venezuela, which exhibited unexpectedly high domestic streaming relative to its peers. Several countries such as Australia, Canada, Belgium, Switzerland, New Zealand, and United Arab Emirates also showed high levels of international streaming. This could be explained by these countries having high English proficiency and stronger exposure to Western media, leading to greater consumption of international English artists.

Finally, the network web plot further revealed country to country interactions and clear clustering patterns (Figure 3). Clear visual clusters included Spanish speaking countries, Eastern European countries, Asian countries, and Western European countries. Borders between countries could also be identified such as Brazil and Portugal being on the border between the Western European and Spanish speaking countries. Countries located near the center tended to be more globally connected and included either English-speaking nations or English proficient nations such as the US, UK, Australia, Ireland, Austria, Sweden, and

the Philippines. One of the biggest players in the plot was Puerto Rico, with many countries streaming from there, despite having a low amount of users and population.

### 3.2 Regional Top 50 Charts Analysis

We first examined the overall distribution of song durations across all selected regions, as shown in Figure 4. The data reveals that the average track length is approximately three and a half minutes. When observing the spread of the data, the majority of the songs—specifically those falling within one standard deviation from the mean—have durations ranging between 160 seconds and 270 seconds. Despite this clear central tendency, the overall distribution of song durations is notably broad. The charts accommodate a wide variety of track lengths, featuring exceptionally short songs that clock in at under one and a half minutes, as well as longer tracks that exceed six minutes in duration. This wide range suggests that global listeners are highly receptive to varying song lengths depending on the track’s appeal.

To further investigate regional music preferences, we utilized both box plots (Figure 5) and density plots (Figure 6) to visualize the distribution of song durations across different countries. The data reveals significant regional variations in listening habits. For instance, the distribution for India is notably broad, characterized by a wider interquartile range in the box plot and a flatter curve in the density plot. This suggests that the Indian market does not exhibit a strong preference for a specific track length. In contrast, Italy’s distribution is exceptionally narrow and concentrated on shorter tracks, indicating a clear preference for shorter song formats in the Italian market.

To investigate whether short song durations correlate with higher chart rankings, we grouped the Top 50 rankings into bins of ten (e.g., ranks 1-10, 11-20) and calculated the deviation of each bin’s average duration from the respective region’s overall Top 50 average (Figure 7). The results demonstrate that the relationship between chart ranking and song duration varies across different regions. For example, in Austria, the highest-ranking tracks (bins 1-10 and 11-20) are shorter than the country’s overall average. Conversely, Mexico’s top 20 positions are primarily occupied by songs that are longer than the regional average. While these visual observations suggest distinct regional trends, we employed rigorous statistical testing to provide a more formal assessment. As presented in Table 3, we conducted a Spearman’s rank correlation analysis to evaluate the relationship between chart ranking and song duration. The results reveal that there is no statistically significant correlation between ranking and track length across any of the examined regions ( $p > 0.05$ ).

Turning our attention to the presence of explicit content, Table 4 reveals that only the United States, the Global chart, and Taiwan feature tracks with explicit labels in their Top 50, while all other countries report zero explicit songs. This stark contrast can likely be attributed to two primary factors. First potential factor is the regional prevalence of genres like hip-hop and rap, particularly in the US, which frequently feature mature themes compared to other markets. Second, this disparity likely stems from the entrenched

US industry standard of applying the "Parental Advisory Label" (PAL).

Finally, we examined the age of the charted songs to determine whether specific regions exhibit a stronger preference for newer releases over older tracks. As illustrated in Figure 8, when using the Global market distribution as a baseline, regions such as Brazil, Italy, and Mexico display a higher concentration of recently released tracks. This heavily right-skewed distribution suggests a strong market preference for contemporary music in these countries. Furthermore, we also investigated the presence of enduring classic tracks on the charts. Notably, we identified three songs released over 40 years ago that remarkably maintain positions on the Global, US, and Austria charts. Upon further investigation, the resurgence of these old tracks can largely be attributed to contemporary viral trends, including prominent placements in popular television series, internet memes, and widespread TikTok challenges.

### 3.3 Lyrical Analysis

First, the LDA topic modeling identified five main lyrical themes in the Spotify US Top 198 songs. These topics were interpreted as romantic emotion and separation, summer and nighttime mood, nightlife and club atmosphere, drinking and emotional reflection, and heartbreak and sadness. As shown in Table 5, the topic distribution is relatively balanced across the corpus, with the first four topics each accounting for around 20% to 25% of the songs, while the heartbreak and sadness topic appears less frequently.

The visualization of the LDA model is shown in Figure 9. The intertopic distance map illustrates the relationship between the five topics in a two-dimensional space. Each circle represents one topic, and the distance between circles reflects how different the topics are from each other. From the figure, the topic clusters are relatively separated, suggesting that the model can distinguish different lyrical themes reasonably well.

In addition, the most salient words associated with the topics are shown on the right side of the figure. Many of these words are related to emotions, relationships, or personal experiences, such as "love", "friend", and "lonely". This indicates that emotional expression plays an important role in popular music lyrics.

Second, sentiment analysis showed that positive lyrics are more common than negative ones. As summarized in Table 6, 130 out of 198 songs were classified as positive, 67 as negative, and only 1 as neutral. This indicates that although many songs include themes such as heartbreak or reflection, the overall emotional tone of popular songs tends to be positive or expressive rather than purely negative.

This pattern is also supported by the figure results. Figure 10 shows that the average compound sentiment score is positive for all ranking groups. The Top 50 songs have an average score of about 0.376, the 51–100 group has about 0.298, the 101–150 group has a lower score of about 0.122, and the 151–200 group has the highest score at about 0.440. Although the 101–150 group appears slightly lower than the other groups, the overall differences are not very large.

In addition, Figure 11 shows that positive songs make up the largest share in every ranking group. Around 70% of the Top 50 songs are positive, 65.3% in the 51–100 group are positive, 55.1% in the 101–150 group are positive, and 72.0% in the 151–200 group are positive. The 101–150 group has a relatively higher proportion of negative songs, but the general pattern remains similar across all groups. Overall, these results suggest that positive emotional expression is a common feature of popular songs, regardless of chart ranking.

We also examined sentiment patterns across different chart ranking groups. Figure 10 shows the average compound sentiment score for each rank group. Overall, all four groups have positive average scores, which means that the emotional tone of the lyrics is generally positive. The Top 50 songs have an average score of 0.376, the 51–100 group has 0.298, the 101–150 group has the lowest score at 0.122, and the 151–200 group has the highest score at 0.440. Although the 101–150 group is slightly lower than the others, the difference across groups is not very large. This suggests that songs at different chart positions still share a similar emotional tone overall.

Figure 11 shows the sentiment composition in each rank group. In every group, positive songs take the largest proportion. For example, positive songs account for 70.0% of the Top 50 group, 65.3% of the 51–100 group, 55.1% of the 101–150 group, and 72.0% of the 151–200 group. The 101–150 group has a relatively higher share of negative songs, but positive songs are still the largest group. This result is consistent with Table 6 and further supports that positive or emotionally expressive lyrics are more common in popular songs.

Finally, keyword and phrase frequency analysis also supports these findings. Table 7 and Figure 12 show the most frequent words in the lyrics corpus. Common words such as “love”, “baby”, “know”, and “like” appear many times, which suggests that popular music lyrics often focus on emotions, relationships, and personal feelings. This result is also consistent with the themes found in the LDA analysis.

Figure 13 and Figure 14 show the most frequent two-word and three-word phrases in the lyrics. Many of these phrases are repeated sounds such as “oh oh”, “la la”, and “yeah yeah”. These phrases do not always carry a strong meaning, but they are often used to create rhythm and repetition in songs. This suggests that many popular songs use simple and repeated phrases to make the lyrics easier to remember and easier for listeners to sing along with. Overall, the keyword and phrase results match the LDA and sentiment results, showing that emotional words and simple repeated patterns are common features in popular music lyrics.

## 4 Summary

### 4.1 Limitations

Due to computation time and rate limits, data acquisition was limited. For example, our methodology for country comparisons were taking the top 20 weekly songs. Just taking



the top songs is not representative of overall listening trends within a country especially if music streams are diverse. Our analysis does not look into within country diversity. For example, a country where top streamed songs have a large dropoff after the top few songs would be fundamentally different from a country that has similar streams from the top songs to medium streamed songs. Thus, increasing the songs would likely yield better results. Initially, we also wanted to look into more countries for the song trend analysis, but Spotify applied heavy rate limits, leading us to pick a select few countries.

Chart trend being limited to only weekly data was a significant limitation. The music industry and streams is severely volatile to quick trends and new music releases, making analyzing weekly charts unreliable. A popular artist that releases a new album in one week could dominate the charts for one week and then drop right after. Our data had Bad Bunny dominating the charts which could likely be influenced by his performance in the 2026 Super Bowl one month prior to our study.

Another limitation was our analysis was focused on Spotify streams only. Drawing conclusions about listener trends can only be done on Spotify users only as they are not a representative sample of the global population. Spotify is an app that biases Westernized countries and younger audiences. For some countries, data is quite limited making any generalizations about listening trends flawed. This was especially true in Eastern European countries such as Belarus, Bulgaria, and Ukraine.

Finally, our data was quite messy. When matching artists to the MusicBrainz database, missing values were found to be one of four reasons: 1. the artists missing from the database, 2. the artist being found but having no associated country/area codes or genres, 3. the database and chart name not matching due to language differences, and 4. name overlaps leading to picking the wrong artist. The latter two reasons could be approached using a better search algorithm such as translations, however the former two are unavoidable given the database. To start, classifying artist genre is problematic as artists seldom classify into one single genre. Genre definitions are also very fluid, with many subgenres and mixtures of genres existing. With the MusicBrainz database being user inputted, the best approach was to take the top genre inputted. Similarly, even artist country is not well defined as country can be defined as to where someone was born in versus where their career is based in. Nonetheless, for our project, we proceeded with the country and genre we obtained and performed manual imputation as previously mentioned in section 2.1.

As stated before, foreign languages, especially Spanish consisted a large portion of the lyrics. Even after subsetting for only US charts, Spanish words were still prominent leading us to manually include Spanish words as stop words.

## 4.2 Conclusions and Future Work

We find that while the US and English speaking nations dominate the historical charts, many other countries are becoming major players in the music industry such as Latin American countries. These trends are also reflected in preferred genre with hip-hop and

pop leading the historical chart while reggaeton has become prominent recently. In addition, we see that there is large variations between countries in terms of artist locality, showcasing strong local music markets or cultural preferences. For song trend, we find that duration distribution differences in countries but is not overall reflective of rank. We also see that while charts bias newer songs, old song outliers are prevalent due to online trends. Lyrical analysis reflects distinct themes reflected in hip-hop and pop genres such as clusters sad, romantic, explicit, and upbeat songs.

One aspect to explore would be to perform the same analysis but across different music streaming platforms such as Youtube, Apple music, and country specific platforms such as QQ music in China. Similar results could indicate generalizability of our Spotify sample while significantly different results could lead to inferences regarding a Spotify specific population. Integration of translations for lyrics would allow us to analyze lyrics for different countries, providing valuable information regarding user preferences. If week by week charts are available, aggregating data to a monthly or yearly level would help in truly analyzing how trends change over time. Finally, a lot of work can be done in genre analysis. If given song metadata, we could potentially implement machine learning methods to predict genre for missing values. Clustering methods would also provide a much more robust analysis regarding across country trends.

## 5 References

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## 6 Figures and Tables

Table 2: Number of Artists in the Top 500 Global Streams for the Top 11 Countries

| Country        | Number of Artists |
|----------------|-------------------|
| United States  | 245               |
| United Kingdom | 58                |
| Puerto Rico    | 31                |
| Mexico         | 27                |
| Canada         | 18                |
| Colombia       | 14                |
| India          | 14                |
| Brazil         | 13                |
| South Korea    | 11                |
| Australia      | 9                 |
| Argentina      | 9                 |

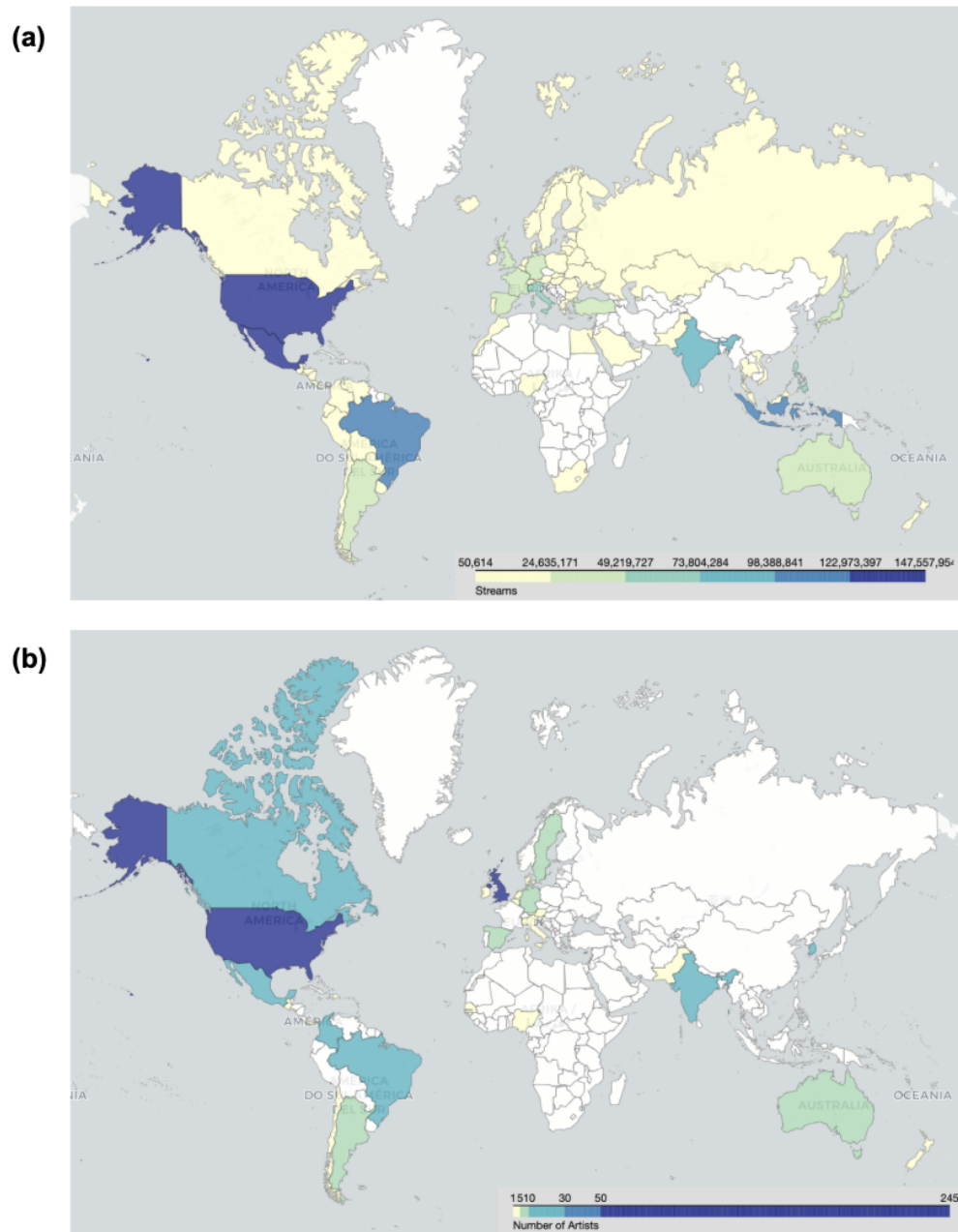


Figure 1: a) Stream counts of top weekly 20 songs for each country (Map). b) Number of artists in top 500 global streams (Map).

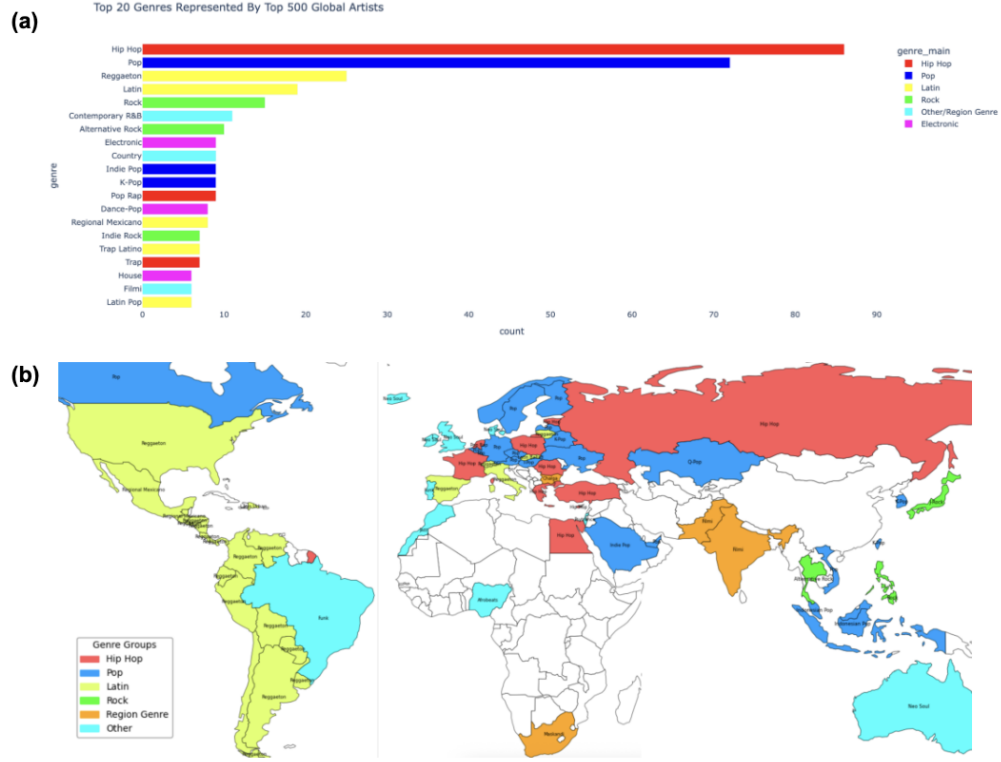


Figure 2: a) Top 20 genres represented by top 500 global artists of all time (Bar Plot). b) Countries by most streamed genre in top 20 songs (Map).

Table 3: Spearman's Rank Correlation Between Chart Ranking and Song Duration

| Country       | Spearman's $\rho$ (Rho) | P-value | Significant? |
|---------------|-------------------------|---------|--------------|
| Global        | 0.08                    | 0.60    | No           |
| Austria       | 0.19                    | 0.19    | No           |
| Brazil        | -0.05                   | 0.74    | No           |
| India         | 0.01                    | 0.93    | No           |
| Indonesia     | 0.13                    | 0.37    | No           |
| Italy         | -0.11                   | 0.47    | No           |
| Japan         | -0.06                   | 0.66    | No           |
| Mexico        | 0.04                    | 0.79    | No           |
| Taiwan        | 0.08                    | 0.60    | No           |
| United States | 0.20                    | 0.17    | No           |

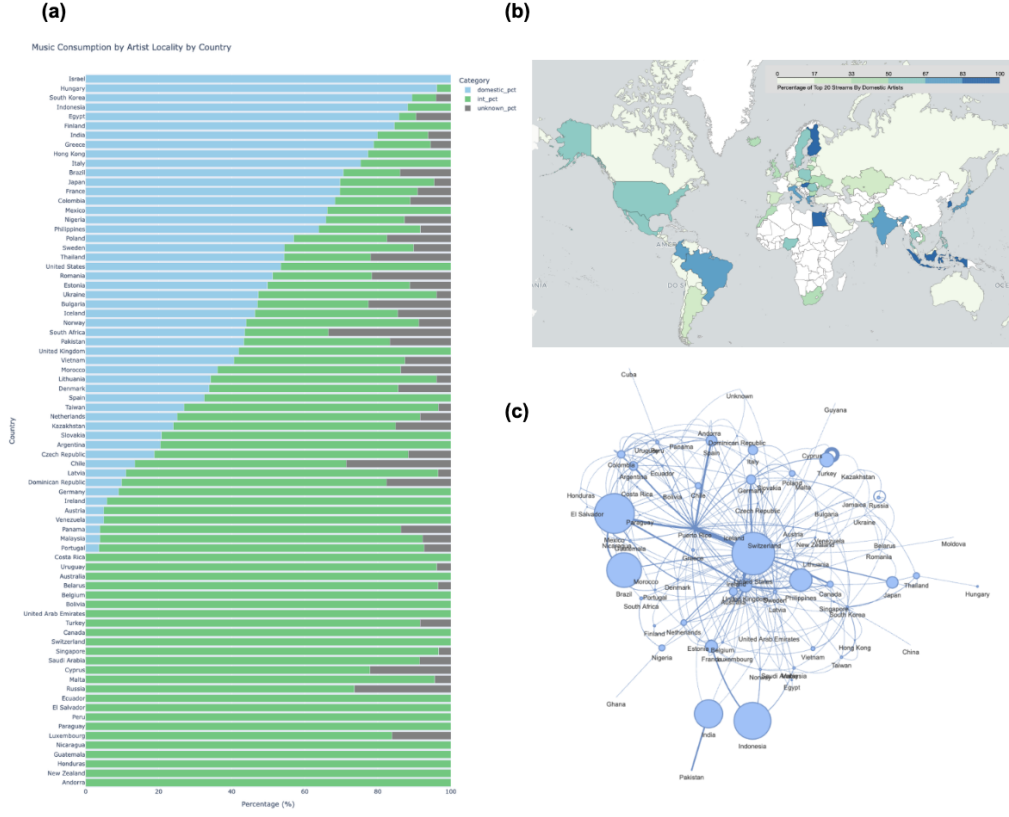


Figure 3: a) Music consumption by percent of streams by domestic, international, and unknown artist (Bar Plot). b) Music consumption by percent of streams domestic (Map). c) Web graph by total streams between two countries. Size of node indicates total stream count and line thickness is higher stream interaction.

Table 4: Comparison of Explicit Content Across Different Country

|   | Country         | Explicit Count | Explicit Percentage |
|---|-----------------|----------------|---------------------|
| 1 | United States   | 16 / 50        | 32%                 |
| 2 | Global          | 11 / 50        | 22%                 |
| 3 | Taiwan          | 2 / 50         | 4%                  |
| 4 | Other Countries | 0 / 50         | 0%                  |

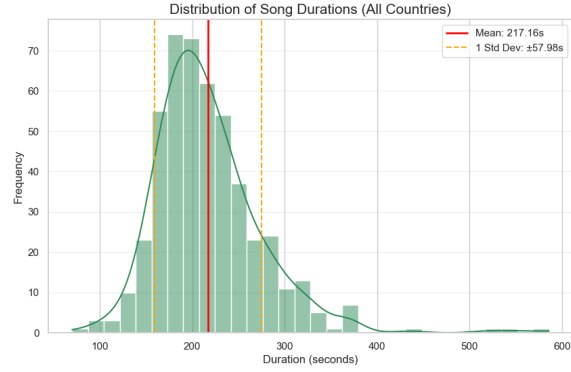


Figure 4: Distribution of Song Durations across Top 50 Charts

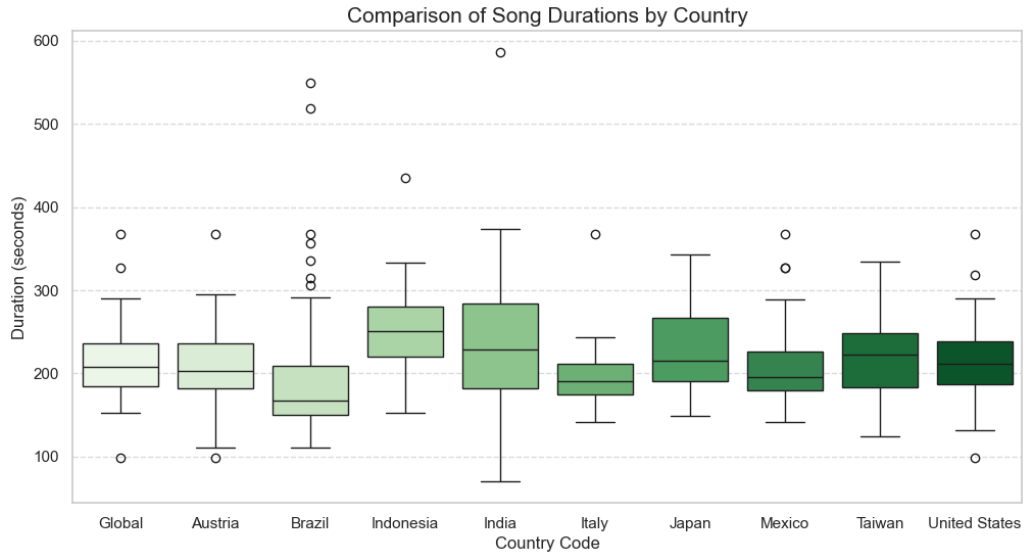


Figure 5: Comparison of Song Durations by Country (Box Plot)

| Table 5: LDA topic distribution for Spotify US Top 198 lyrics |                                   |       |            |
|---------------------------------------------------------------|-----------------------------------|-------|------------|
| Topic                                                         | Interpreted Theme                 | Count | Percentage |
| 0                                                             | Romantic emotion and separation   | 48    | 24.2%      |
| 1                                                             | Summer and nighttime mood         | 45    | 22.7%      |
| 2                                                             | Nightlife and club atmosphere     | 41    | 20.7%      |
| 3                                                             | Drinking and emotional reflection | 40    | 20.2%      |
| 4                                                             | Heartbreak and sadness            | 22    | 11.1%      |

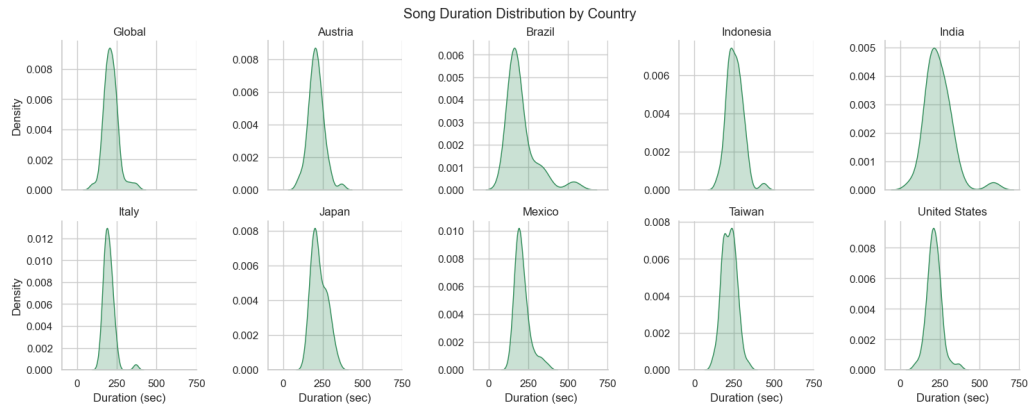


Figure 6: Song Duration Distribution by Country (Density Plot)

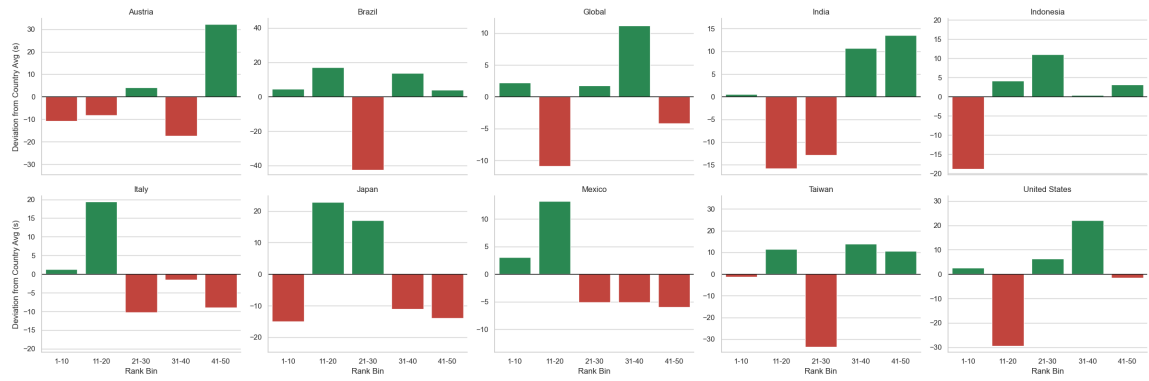


Figure 7: Deviation of Average Song Duration from Regional Mean by Rank Bin

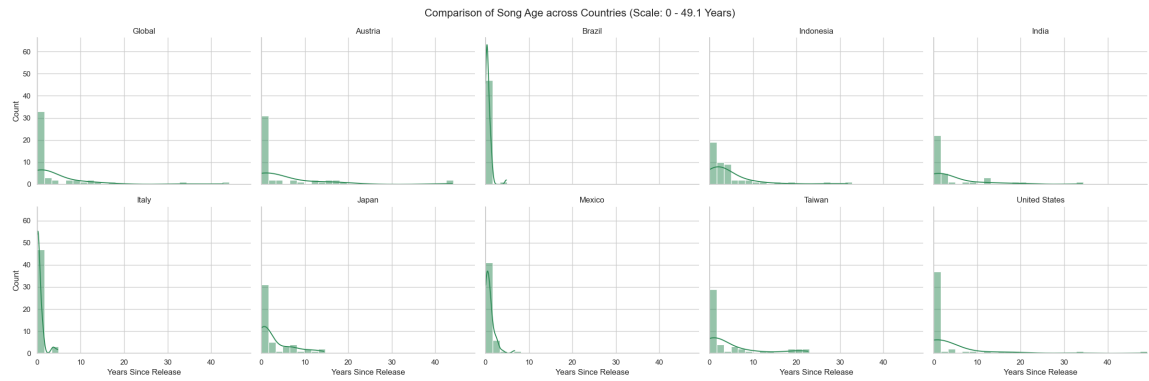


Figure 8: Distribution of Song Age Across Countries



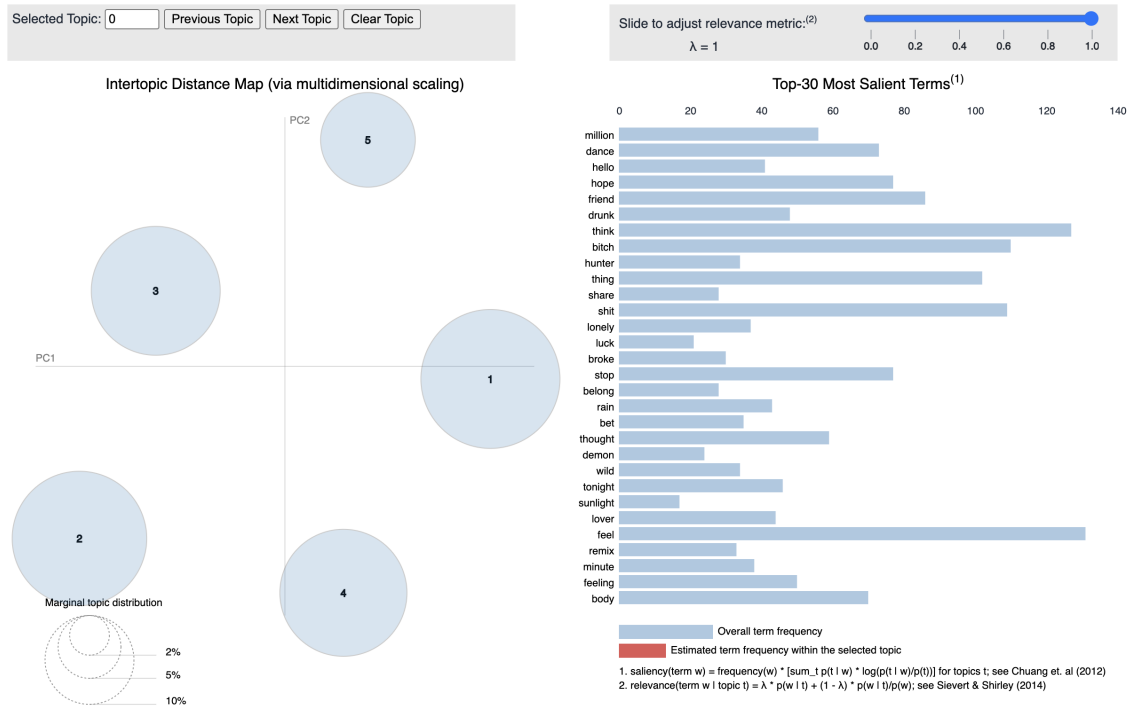


Figure 9: LDA topic visualization using LDAvis. The left panel shows the intertopic distance map, and the right panel shows the most salient words across topics.

Table 6: Sentiment distribution of lyrics in the Spotify US Top 198 songs

| Sentiment | Count | Percentage |
|-----------|-------|------------|
| Positive  | 130   | 65.7%      |
| Negative  | 67    | 33.8%      |
| Neutral   | 1     | 0.5%       |

Table 7: Most frequent unigrams in the lyrics corpus

| Word | Frequency |
|------|-----------|
| love | 120       |
| like | 115       |
| know | 110       |
| baby | 102       |
| yeah | 97        |

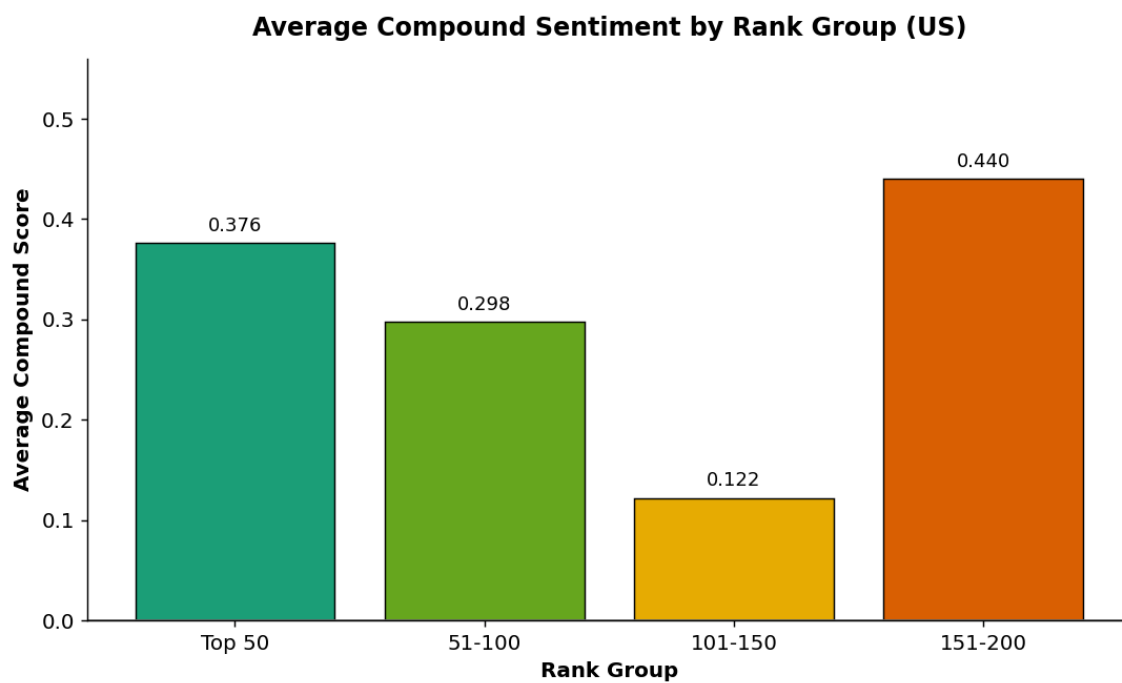


Figure 10: Average compound sentiment score by Spotify rank group

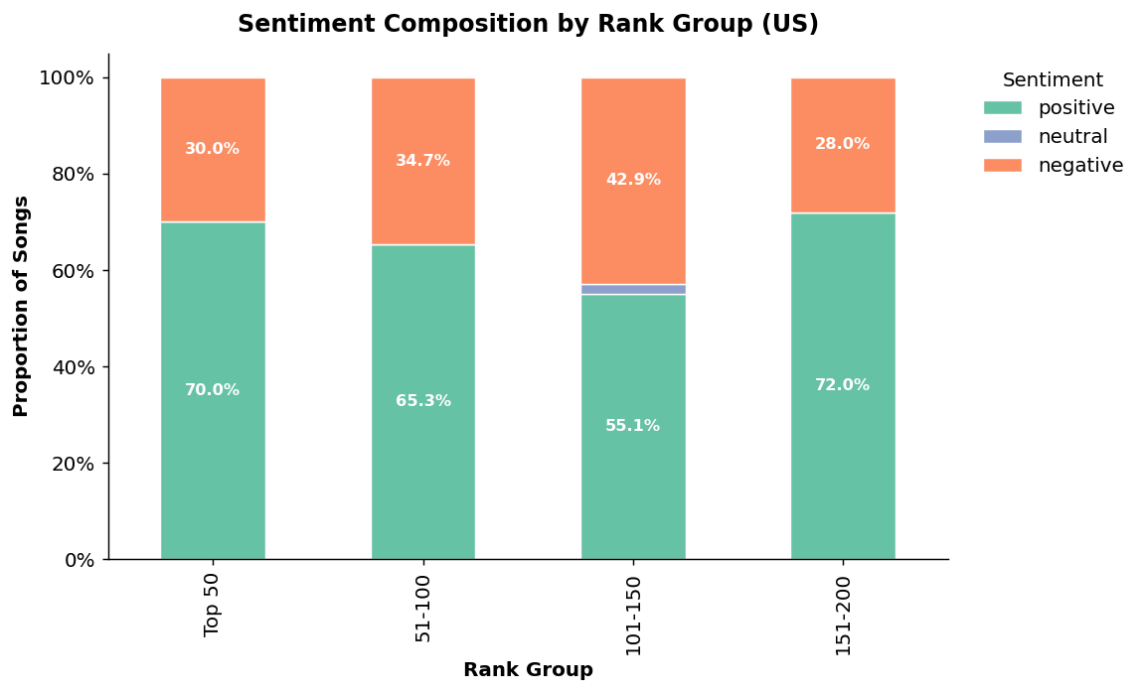


Figure 11: Sentiment composition of songs across different Spotify rank groups

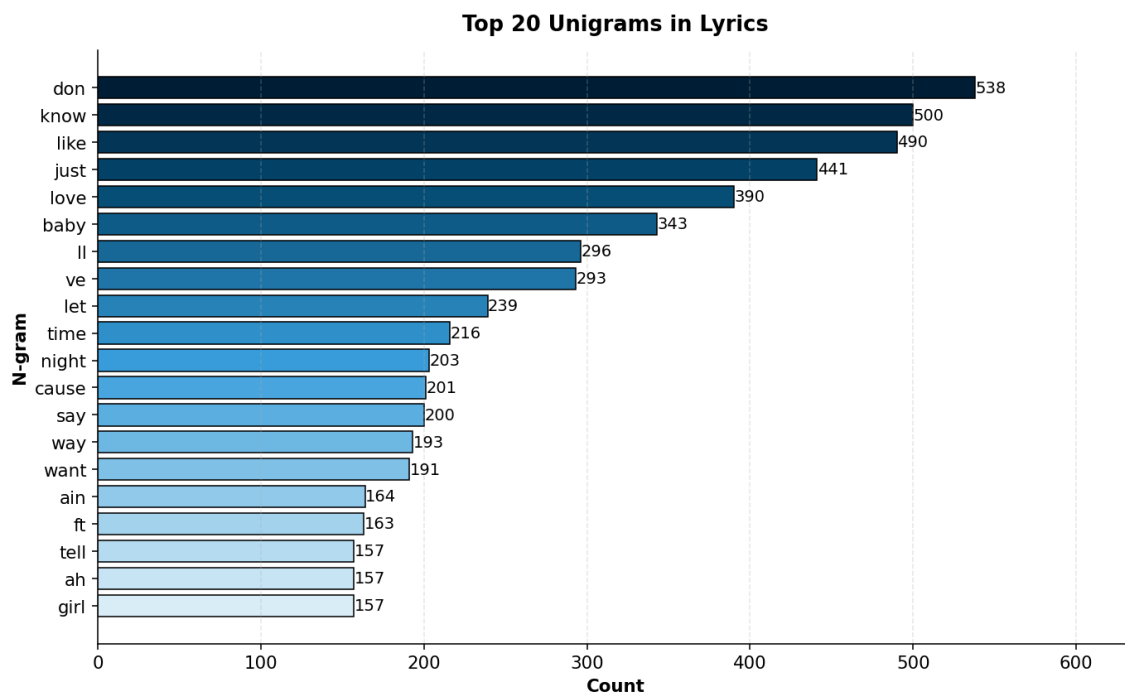


Figure 12: Top 20 most frequent unigrams in the lyrics corpus

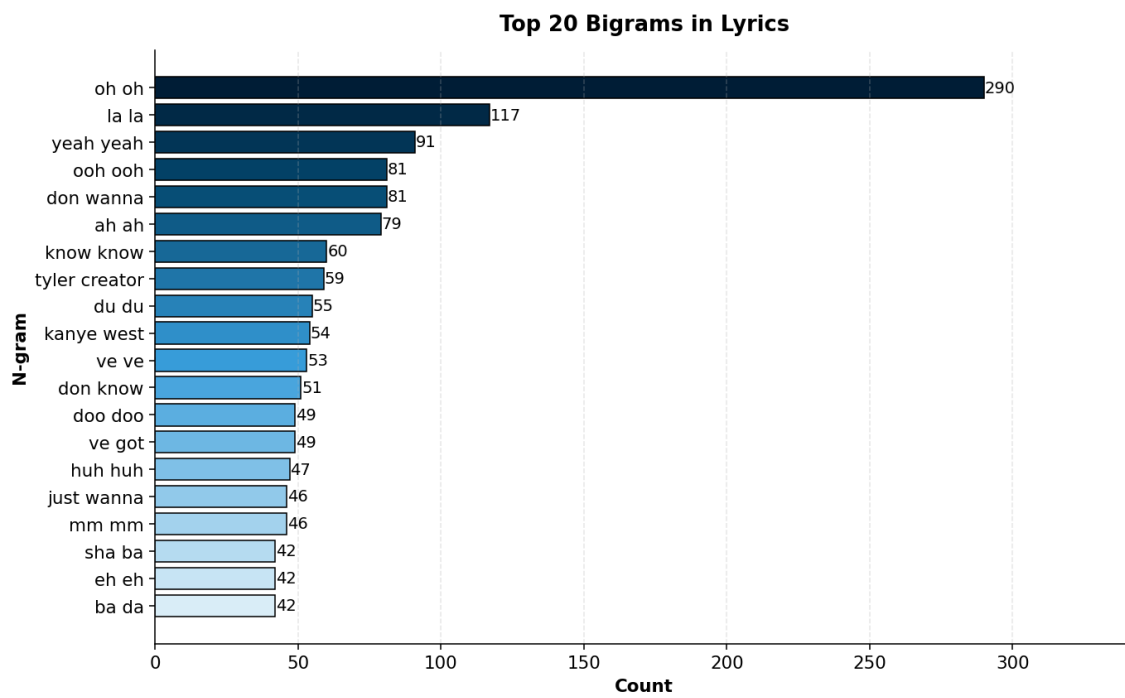


Figure 13: Top 20 most frequent bigrams in the lyrics corpus

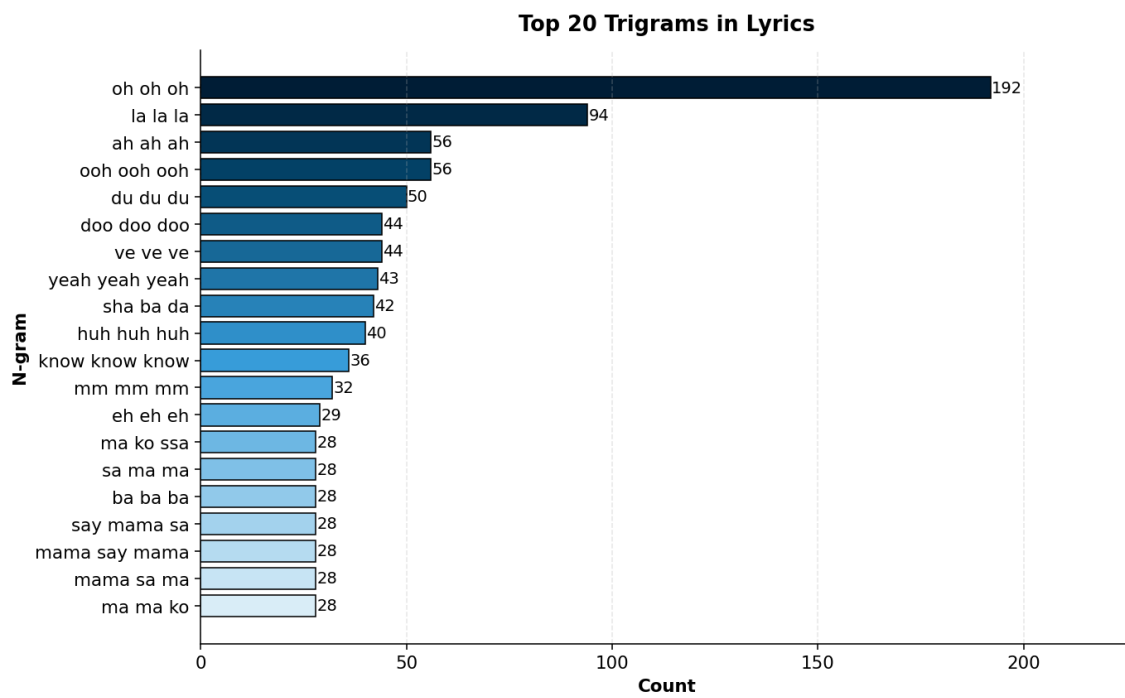


Figure 14: Top 20 most frequent trigrams in the lyrics corpus